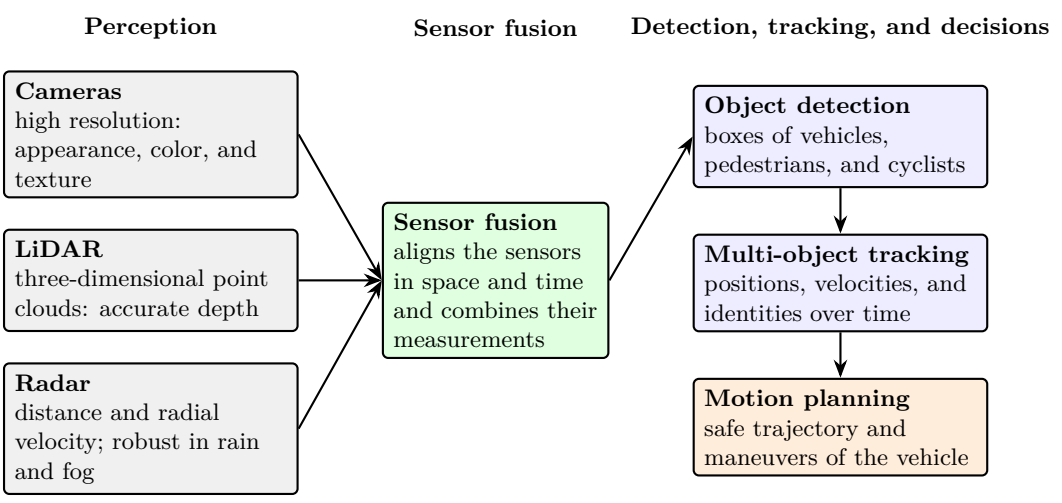

Perception
Sensor fusion
Detection, tracking, and decisions
Cameras
high resolution: appearance, color, and texture
LiDAR
three-dimensional point clouds: accurate depth
Radar
distance and radial velocity; robust in rain and fog
Sensor fusion
aligns the sensors in space and time and combines their measurements
Object detection
boxes of vehicles, pedestrians, and cyclists
Multi-object tracking
positions, velocities, and identities over time
Motion planning
safe trajectory and maneuvers of the vehicle